



New design strategies of deep heterogeneous convolutional neural networks ensembles for breast cancer diagnosis

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Abstract

One of the most consequential public health issues in the world and a major factor in women's mortality is breast cancer. Early detection and diagnosis can significantly improve the likelihood of survival. Therefore, this study suggests a deep end-to-end heterogeneous ensemble approach by using deep convolutional neural networks models for breast histological images classification tested on the BreakHis dataset. The proposed approach showed a significant increase of performances compared to their base learners. Thus, seven deep learning architectures (VGG16, VGG19, ResNet50, Inception V3, Inception ResNet V2, Xception, and MobileNet V2) were trained using fivefold cross-validation. Thereafter, deep end-to-end heterogeneous ensembles of two up to seven models were constructed based on three selection criteria's (by accuracy, by diversity, and by both accuracy and diversity) and combined with two voting methods: majority voting by tacking the mode of the distribution of the predicted labels, and weighted voting by taking the average of predicted probabilities. Results showed the effectiveness of deep end-to-end ensemble learning techniques for histopathological breast cancer images classification since the ensembles designed using weighted voting with the selection by accuracy strategy method exceeded the ones designed using the selection by diversity or by accuracy and diversity strategies. The accuracy values of the proposed approach have shown a significant amelioration compared to the least performing base learner used as a baseline ResNet 50 with an accuracy increased from 78.14%, 78.57%, 82.80 and 79.43% to 93.8%, 93.4%, 93.3%, and 91.8% through the BreakHis dataset's four magnification factors: 40X, 100X, 200X, and 400X respectively.

Keywords Machine learning · Deep learning · Classification · Ensemble learning · Image processing · Digital pathology

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1 Introduction

Cancer is considered among the most consequential health issues in the world. In 2020, more than 19.3 million new cancer cases are diagnosed and nearly 10 million deaths are declared [1]. By far the most eminent and leading cause of death in women worldwide is breast cancer (BC), with 2.3 million women affected by it in 2020 [1]. Early diagnosis of BC is essential to minimize morbidity in women. Even though X-ray, MRI (Magnetic Resonance Imaging), ultrasound, and other imaging techniques have been used for more than 40 years to detect breast cancer [2], biopsy techniques have always been the most commonly used method for correctly diagnosing breast cancer. Gathering tissue samples, mounting them on microscopic glass slides, and staining them for visualization are all part of the procedure [3]. To confirm the diagnosis, pathologists examine and diagnose the histopathological images [3]. Manual examination of large-scale histological images, however, is a difficult process due to changes in appearance, structure, and textures [4], it is time-consuming, and usually depends on human subjective interpretation since the level of experience of the pathologists involved may influence the results of the analysis. Therefore, computer-aided [5] analysis of histological images are crucial in the diagnosis of breast cancer. Deep learning (DL) has recently outperformed a variety of machine learning (ML) models for the medical image analysis tasks, such as classification [6, 7], detection [8], and segmentation [9]. When compared to other types of ML classifiers, DL has the advantage of being able to achieve results that are similar or better than human performance. DL techniques have been used in computer vision [10], biological science [11–13], and many other domains to solve problems of traditional feature extraction. Deep convolutional neural networks (DCNNs) in particular have been widely recognized as one of the most useful techniques for image classification as they offer distinct advantages over conventional solutions, including an end-to-end architecture that frees users from hand-crafted feature extraction tasks [14]. Regardless of how popular it is, a single DCNNs model can only extract a limited number of discriminative features, resulting in poor overall classification performance. To improve classification accuracy, DCNNs architecture ensembles have been created to discover the representation of histopathological images from various perspectives.

Since then, many researchers started investigating deep ensemble learning techniques to alleviate the performances [15–19]. Although ensemble learning have known an increase of use in the medical domain especially in BC [20], it is observed that most researchers design ensembles using only one strategy based on combining weak learners and without taking into consideration the diversity which is a key element to design well performing ensemble learning models by taking into consideration the differences or variations among the individual models that constitute an ensemble [21, 22]. For instance, the study [22] proposed heterogeneous ensembles for BC classification using three datasets (WDBC, Wisconsin, WPBC). From seven classifiers (KNN, Decision Tree, MLP, SVM, SVM-PUK, SVM-RBF, S-LK, SVM-NP), the authors, selected and constructed ensembles based on two selection strategies: (1) selection by accuracy and diversity, and (2) selection by accuracy. Then, the constructed ensembles were combined using the majority voting method. According to the outcomes of this study, investigating both criteria's accuracy and diversity to select base learners for the proposed ensembles often resulted in greater performance than designing and building ensembles without member selection. In [23], authors proposed an ensemble constructed with three DL models for breast cancer classification over the BreakHis and Bioimaging-205

datasets. Inception ResNet V2 and gradient boosting trees were used to extract features and classification respectively. The classifiers were combined using the majority voting strategy to improve their performances. The accuracy values reached: 95.1%, 96.3%, 96.9%, 93.8% and 86.75% for the four magnification factors (MF) of the BreakHis dataset and bioimaging dataset respectively. Some limitations have been revealed in the studies [23, 24]: (1) the design of the proposed ensembles used only one combination method, (2) the design of the proposed ensembles used only one selection strategy, and (3) except of the study [22], a lack of statistical analysis to select the outperforming proposed model.

To elevate the burden of those limitations, this study proposes a deep end-to-end heterogenous ensemble (DEHtE) approach using deep learning models as base learners for a binary classification of breast histopathological images over the BreakHis dataset by investigating three selection strategy in the design of the proposed ensembles. The proposed approach consists of combining seven DL techniques using two voting methods: majority voting by taking the mode of the distribution of the predicted labels, and weighted voting by taking the average of predicted probabilities. The seven DL techniques (VGG16, VGG19, ResNet 50, Inception V3, Inception ResNet V2, Xception, and MobileNet V2) were pre-trained on ImageNet, loaded and trained using a fivefold cross-validation. From the predictions of the seven DL models, we construct ensembles of two up to seven members based on three selection criteria's (by accuracy, by diversity, and by both accuracy and diversity) in order to investigate which is the best strategy in the design of the proposed ensembles. The performances were then evaluated using four classification performance measures [25, 26] (accuracy, precision, recall, and F1-score), Scott Knott (SK) statistical test to group the proposed ensembles and identify the best cluster, and the Borda Count voting method to sort and identify the best performing ensemble. In light of what we know, this study is the first to design a new approach consisting of deep end-to-end heterogenous ensemble (DEHtE) using DL models as base learners based on three selection strategies (by accuracy, by diversity and by both accuracy and diversity), and combined with two voting methods (majority and weighted voting) for histopathological BC classification.

The current study focuses on three research questions (RQs):

- (RQ1): Does the DEHtE using voting combination methods outperform the DL models used as base learners?
- (RQ2): Which selection strategy in the design of the proposed approach (DEHtE) gives better results?
- (RQ3): Which voting combination method gives better results for the design of the DEHtE?

The main contributions of this study are the following:

1. Designing DEHtE using DL models as base learners with three selection strategies: by accuracy, by diversity and by both accuracy and diversity.
2. Combining the proposed DEHtE using majority and weighted voting methods.
3. Assessing and comparing the performance of the proposed new approach: DEHtE with their base learners over the BreakHis dataset.
4. Selecting which strategy, combination method and number of base learners gives better results in the design of the DEHtE.

The remainder of this paper is structured as follows: Sect. 2 presents an overview of some selected studies related to this work. Section 3 briefly describes the deep learning models used, the concepts of ensemble learning, and the Q measure used to calculate the diversity between the DL models. Section 4 presents data preparation process. Section 5 provides the details of the experiment configuration, the empirical methodology and the abbreviations followed in this empirical study. Section 6 reports and discusses the empirical results. Section 7 covers the threats of validity of this study. Lastly, Sect. 8 outlines the conclusion and ongoing work.’

2 Related work

The primary outcomes of the research conducted on ensemble learning for BC diagnosis are outlined in this section. To elaborate, we initially discuss the key findings from recent studies that have assessed the application of DL architectures in the realm of digital pathology. Following that, we detail the significant results derived from practical studies that have introduced and assessed ensemble learning methods for the purpose of diagnosing breast cancer.

Table 1 describes the main outcomes of six relevant papers dealing with ensembles learning strategies for BC classification [15, 27–31]. The following are observed:

- Most of the studies focused on a single combination method for constructing their proposed approaches. Hence, introducing a comparison of different combination strategies would an added value to the existing literature.
- Most of the studies take into consideration the selection strategy accuracy, and no other study investigated other selection strategies to combine the base learners.
- Despite the dataset’s limited size, many studies employ data splitting as a validation technique. Therefore, the use of cross validation is highly recommended to validate the performances of the proposed models since it provides stronger and more dependable performance estimations by making better use of the data, diminishing variability, and offering a thorough evaluation of model generalization.
- A small number of studies constructed deep ensemble learning models employing more than three DL models.
- The accuracy is the most used metric for the evaluation of the proposed approaches.
- Lack of use of statistical tests to evaluate the proposed approaches.

3 Background

This section delves into some of the key principles used in this empirical study by discussing ensemble learning techniques and the diversity measure used in this paper.

3.1 Ensemble learning

Ensemble Learning was originally proposed for classification tasks in 1965 [32], It is based on the concept of training multiple base learners as ensemble members and

Table 1 Overview of six relevant papers proposing deep ensemble learning models for BC classification

Authors and year of publication	Base Learners	Ensemble Methods	Combination Method	Dataset	Findings	Limitations
[27], 2023	VGG16, Xception, ResNet 50 and DenseNet 201	Heterogenous Ensemble	Weighted voting	BreakHis dataset	Propose an ensemble learning model by combining the four best performing DL architectures: ResNet 50, Xception, VGG16 and DenseNet 201 trained on the BreakHis dataset on Image level regardless the MF used and using transfer learning, weighted voting as a combination method and based on accuracy as a selection strategy. Results proved the potential of the proposed approach since it achieved an accuracy of 98.90%	-Use of data splitting (Train, validation and test sets) instead of K-fold cross validation -Use of only one selection strategy to design the ensembles -Use of one combination method -Lack of use of statistical test for the selection of the best performing method -Train the ensemble on Image level regardless the MF used
[28], 2023	ResNet 50 and SVM	Heterogenous ensemble	-	BreakHis dataset	Propose an ensemble approach by using K-means for image segmentation, ResNet 50 for feature extraction and SVM for classification tested using the BreakHis dataset. The findings advocate the proposed approach since it achieved an accuracy value of 92.20%, 90.60%, 92.60% and 89.7% for the MFs 40X, 100X, 200X and 400X respectively	-Use of a hybrid architecture rather than an ensemble learning approach -Use of data splitting for the model evaluation rather than K-fold cross validation -Lack of use of statistical tests to evaluate the proposed approach

Table 1 (continued)

Authors and year of publication	Base Learners	Ensemble Methods	Combination Method	Dataset	Findings	Limitations
[29], 2023	AlexNet, VGG16, ResNet34, ResNet50, DenseNet201	Heterogenous Ensemble	Histo-Fusion Strategy	BreakHis Dataset	Propose an ensemble model using histo-fusion method by training the DL models on high resolution images and then fuse them with low resolution images. As indicated by the findings of this study, the Histo-Fusion learning method that was introduced has enhanced the discriminative capabilities of each deep convolutional neural network. This enhancement has led to an increase in accuracy scores of approximately 5% compared to the accuracy achieved through the widely employed transfer learning approach	-Use of data splitting method for validation rather than cross validation - Lack of use of statistical tests to evaluate the proposed approach -Lack of investigation of the use of diversity -No use of combination methods but rather use of a fusion strategy
[30], 2021	Inception V3, Inception ResNet V2 and Xception	Heterogenous Ensemble	Concatenation of the features extracted	BreakHis Dataset	Develop a new approach using DL architecture by fusing features and testing them on the BreakHis dataset. The authors tested many combinations of top two DL models, and as results the best proposed ensemble is the one fusing the features of Xception and Inception V3 with an accuracy value reaching 97.5%	-Use of data splitting instead of cross validation -Use of concatenation instead of a combination rule -Lack of use of statistical test to compare the proposed models -Lack of investigating different combination and selection strategies

Table 1 (continued)

Authors and year of publication	Base Learners	Ensemble Methods	Combination Method	Dataset	Findings	Limitations
[15], 2023	VGG16, VGG19, ResNet50, Inception V3, ResNet V2, Xception and MobileNet V2	Heterogeneous Ensemble	Stacking using Logistic regression as a meta learner	BreakHis dataset	Propose a deep heterogeneous ensemble by combining seven DL models for both feature extraction and classification and combining their prediction following the stacking strategy by using logistic regression as a meta learner. Results advocates the use of the proposed approach since it outperformed its base learners and achieved an accuracy of 93.8%, 93.00%, 93.30% and 01.8% for 40X, 100X, 200X and 400X respectively	-Lack of investigation of different selection strategies -Use of one ensemble learning strategy -Lack of comparison of the proposed ensemble with other ensemble learning strategies
[31], 2023	ResNet50 and AlexNet for Feature extraction K*, Logistic Discrimination, Naïve Bayes, Random Forest and SVM	Heterogeneous Ensemble	Accuracy averaging	The Center of Bio-image Informatics, University of California, Santa Barbara (UCSB) dataset	Propose a hybrid ensemble model by selecting and combining the most relevant features using DL models (AlexNet and ResNet50) and feeding them to an ensemble of classifiers (K*, Logistic Discrimination, Naïve Bayes, Random Forest and SVM). Results after testing the proposed approach shows its superiority by reaching an accuracy of 98%	-Lack of use of statistical tests to compare the proposed models -Lac of use of a variety of combination methods -Lack of investigation of different combination strategies

afterwards combining their predictions into a final output that should outperform any other ensemble member with statistically independent error on the target dataset on average [33]. An ensemble contains a number of learners called base learners. A base learner, which can be a decision tree, a neural network, or any other type of learning algorithm, is generally used to generate base learners from training data. Learners from the same type generate homogeneous ensembles, while learners of different algorithms generate heterogeneous ensembles. An ensemble's generalization ability is frequently much stronger than that of base learners.

The current study uses heterogenous ensembles with majority and weighted voting methods to combine predictions of the base learners:

- (1) In majority voting, each classifier votes for one class label, and the final output class label is the one that gets more than half of the votes, furthermore no weights are used to combine the base learners.
- (2) In weighted voting, each classifier probabilities is considered, these probabilities will be weighted and averaged, consequently the winning class will be the one with the highest weighted and averaged probability [33]. In this study we attributed weights for each base learners according to the selection strategy followed: (1) for the selection by accuracy we ranked the base learners based on accuracy and attributed for each base learner the weights according to their ranks, (2) for the selection by diversity we calculated the Q-statics pairwise, combined the one having the best Q-value and attributed the weights following their ranking and (3) for the selection by accuracy and diversity strategy we ranked the base learners using the Borda Count voting method based on accuracy and the Q-statics pairwise.

3.2 Pairwise diversity measures

The concept of diversity plays a crucial role in the effectiveness of ensemble learning. Diversity refers to the differences or variations among the individual models that constitute an ensemble. Having diverse models is important because it contributes to the ensemble's ability to make accurate and robust predictions. A traditional approach to measuring ensemble diversity is to compare the pairwise similarity of two learners [34], and then take the average of all pairwise measurements for overall diversity.

Provided a dataset: $D = \{(x_1, y_1), \dots, (x_m, y_m)\}$, for binary classification: $(y_i \in \{-1, +1\})$, for two classifiers, we have the following contingency table h_i and h_j , as shown in Table 2 where $a + b + c + d = m$ are non-negative variables that represent the number of examples that fulfill the criteria indicated by the corresponding rows and columns. Based on these variables, the Q-statistic pairwise measure was developed.

Q-Statistic [35] is defined as:

$$Q_{ij} = \frac{ad - bc}{ad + bc} \quad (1)$$

Table 2 Contingency table for h_i and h_j classifiers

	$h_i = +1$	$h_j = -1$
$h_i = +1$	a	c
$h_j = -1$	b	d

It can be seen that Q_{ij} takes value in the range of $[-1, 1]$. Q_{ij} is zero if h_i and h_j are independent; Q_{ij} is positive if h_i and h_j make similar predictions; Q_{ij} is negative if h_i and h_j make different predictions.

4 Data preparation

In this section, we will present the process to prepare the histological BreakHis dataset consisting of five steps: data acquisition, data pre-processing using Contrast Limited Adaptive Histogram Equalization (CLAHE), intensity normalization and data augmentation. Since the same process was followed in the studies [36, 37], we therefore summarize it as described below.

BreakHis contains hematoxylin–eosin-stained breast histological slides, which are microscopic examinations of a biopsy to study the appearances of cancer. It is constituted of 7,909 breast histopathological images collected from 82 patients at different magnification factors (MF) such as 40X, 100X, 200X, and 400X with effective pixel sizes of 0.49 m, 0.20 m, 0.10 m, and 0.05 m. All images are stored in TrueColor (24-bit colour depth, 8 bits per colour channel) three channel format (RGB). A pathologist examined the images and identified the most relevant region of interest, excluding out-of-focus images and images with undesirable areas such as black borders or text annotations. Each image's final size is 700×460 pixels in PNG format [38, 39]. The dataset is divided into benign tumor (adenosis, fibroadenoma, phyllodes tumor, and tubular adenoma) and malignant tumor (ductal carcinoma, lobular carcinoma, mucinous carcinoma, and papillary carcinoma) [40]. One of the benefits of the BreakHis dataset is the use of four magnification factors, which allows for the detection of various cancer types and subtypes [41]. We used intensity normalization and CLAHE [37, 42] to improve the image quality. The data augmentation was used to deal with the imbalanced data by incorporating geometric transformations [25, 42] since the number of images in each category non-cancerous (2480 images) and cancerous (5429 images) are imbalanced with 70% of the images represent the malignant class.

5 Empirical design

In this section, the empirical design proposed to build the DEHtE is provided, starting with the process tailored to construct and evaluate the designed proposed approach including the experiment configuration. Next, the abbreviations employed to refer to the designed ensembles and their base learners are given. At last, a framework of the experiment and the empirical design is described.

5.1 Experiment configuration

After the preprocessing step, the data is divided into two sets (train and test) with ratios of (80%, 20%). Then, for each dataset MF, seven DL models were trained for both feature extraction and classification by following the transfer learning strategy as illustrated in Fig. 1. The models were trained on the train set and tested on the test set using stratified K-fold cross validation with $k=5$. The following configurations were followed:

- Except for models from the Inception that were resized to 299×299 , all images were resized to 224×224 pixels.
- For the training, we used the transfer learning technique for fine tuning where the convolutional base of the pre-trained architecture was frozen (ImageNet weights are used) and feed the features extracted to the ANN classifier. The structure of the ANN classifier is: (1) a fully connected layer of 256 neuros with RELU activation function, (2) a dropout layer with rate set of 50%, (3) a dense layer of two neuros and SoftMax activation function.
- The training parameters were set to a batch size of 32 and a number of epochs of 200 epochs. As for the optimization, Adam optimizer was used with a learning rate set to 0.0001 that decreases during the training. Moreover, the L2 regularization was used to reduce the model overfitting.

The proposed architectures for this experiment are trained and tested in Python using the Keras and Tensorflow deep learning frameworks and run on a TPU processing unit with 8 cores, 35 GB of RAM, and a Linux-based OS provided by Google in Colab Notebook. Figure 1 depicts the training process for the seven DL models.

5.2 Experimental methodology and process

Figure 2 illustrates the methodology and the process followed to design the DEHtE flowing the three selection strategies. This experiment consists of the following seven steps:

- **Step 1:** Assessing the performance of the seven deep learning architectures in term of accuracy.
- **Step 2:** Designing for each MF, combinations of 2 up to 7 DL architectures used as base learners (combinations of top 2, 3, 4, 5, 6, and 7) following three selection strategies:
 - 1) **Selection by accuracy (SbA):** Ranking the seven DL techniques based on accuracy then from top to down constructing combinations of 2, 3, 4, 5, 6 and 7.

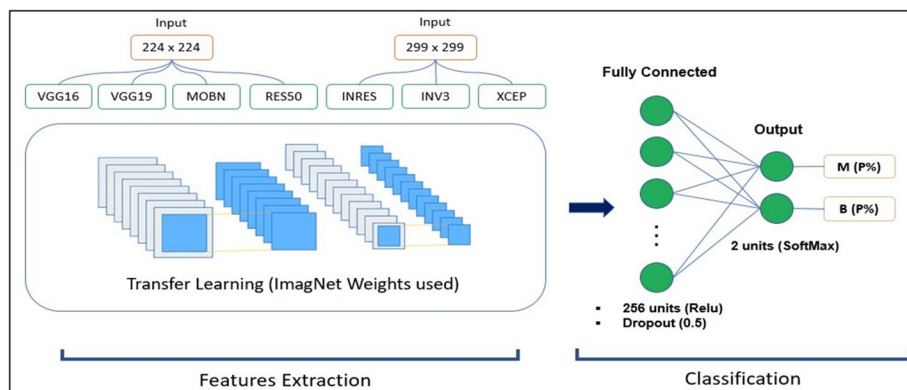
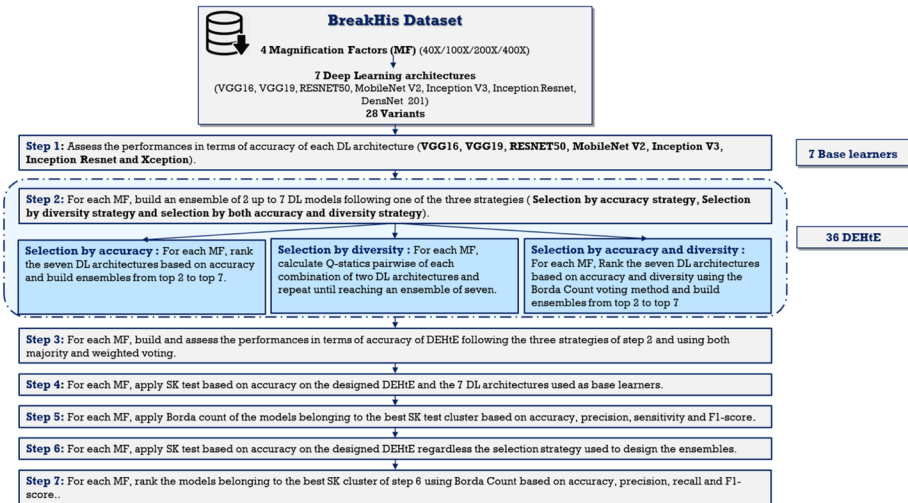
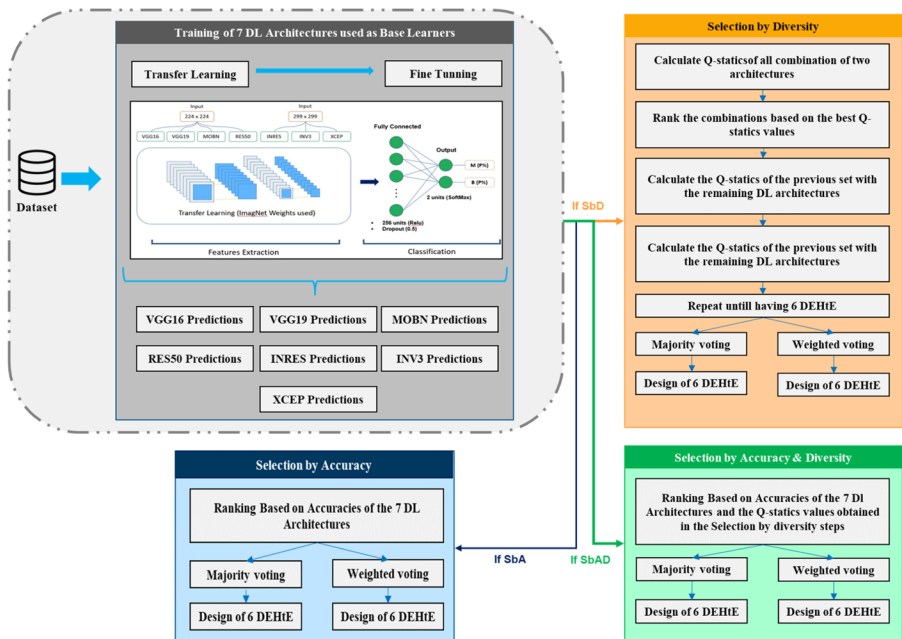


Fig. 1 Training process of the seven DL techniques



(a) Experiment Methodology



(b) Experiment Process

Fig. 2 Representation of the (a) Experiment methodology and (b) experiment process for the design of the DEHiE using three selection strategies (by accuracy, by diversity and by both accuracy and diversity) and two voting methods (majority and weighted voting). Note that the abbreviation used in the Figure are defined in Sect. 5.3

- 2) **Selection by diversity (SbD):** Following the three steps bellow to design the ensembles using the SbD strategy:
 - i. calculate Q-statistic diversity of all combinations of sets of size two and select those that give the best Q value.
 - ii. calculate Q-statistic diversity of all combinations of sets of size two and select those that give the best Q value.
 - iii. calculate Q-statistic diversity of all combinations of sets of size two and select those that give the best Q value.
- 3) **Selection by accuracy and diversity (SbAD):** Ranking the seven DL techniques with Borda Count based on accuracy and diversity, then construct from top to down combinations of 2, 3, 4, 5, 6 and 7.

At the end of this step, we obtain 6 combinations for each selection strategy used (18 combinations in total).

- **Step 3:** For each MF, apply the hard and weighted voting methods on all combinations obtained in step 2, to obtain 36 ensembles in total (3 selection strategies $\times$ 6 DEHtE $\times$ 2 voting methods). Then, evaluate their performance in term of accuracy.
- **Step 4:** For each MF, apply the SK test based on accuracy on each 12-ensembles obtained with each selection strategy in step 3 (6 DEHtE $\times$ 2 voting methods) with the seven DL techniques.
- **Step 5:** This step consists of ranking the variants of the best clusters (obtained in step 4) using Borda Count based on the four-performance metrics.
- **Step 6:** To compare the three selection strategies, apply the SK test based on accuracy on all DEHtE regardless the strategy used (36 ensembles) for each MF.
- **Step 7:** This step consists of ranking the variants of the best clusters obtained in step 6 using Borda Count based on the four-performance metrics.

5.3 Abbreviation

To guide and support the reader and abbreviate the names of the different methods used in this study we have shortened the names of each method as follows: INV3 as Inception V3, INRESV2 as Inception ResNet V2, MOBN as MobileNet, RES50 as ResNet50, and XCEP as Xception

For ensembles names the following abbreviations were chosen:

EnHVA: Ensemble of size n combined with hard voting method and constructed with SbA strategy

EnHVD: Ensemble of size n combined with hard voting method and constructed with SbD strategy

EnHVAD: Ensemble of size n combined with hard voting method and constructed with SbAD strategy

EnWVA: Ensemble of size n combined with weighted voting method and constructed with SbA strategy

EnWVD: Ensemble of size n combined with weighted voting method and constructed with SbD strategy

EnWVAD: Ensemble of size n combined with weighted voting and constructed with SbAD strategy.

6 Results and discussions

This section evaluates and compares the performance of the DEHtE designed using the three selection strategies and their base learners over the four MFs of the BreakHis dataset (40X, 100X, 200X and 400X). Moreover, we identify the suitable selection strategy, number of base learners and voting combination methods. To do that:

- (1) The performances were analyzed and compared based on accuracy of all proposed DEHtE and their base learners; thereafter, we clustered and ranked them using SK test (based on accuracy) and Borda Count voting method (based on accuracy, recall, F1-score and precision) respectively (RQ1), and
- (2) To select the best performing and statistically indifferent proposed approach, the best selection strategy, and the best combination method, SK statistical test was applied based on accuracy on all proposed DEHtE regardless the selection strategy; thereafter, the models belonging to the best SK cluster were ranked using Borda Count voting method (RQ2 and RQ3).

6.1 Assessing the performance of the DEHtE and their base learners

This subsection aims to evaluate and compare the performances of DEHtE constructed based on three strategies: (1) selection by accuracy, (2) selection by diversity and (3) selection by accuracy and diversity. For each selection strategy we applied two voting techniques (majority and weighted voting), and the results were presented based on accuracy. Moreover, to see how ensembles results are significantly different than the seven DL technique used as based learners, the SK statistical test was used, and the methods belonging to the best cluster were ranked using: (1) Borda Count based on four performance measure (accuracy, precision, recall and F1-score) for the selection by accuracy strategy; (2) Q-value calculated using Q-statistics as a diversity measure for the selection by diversity, (3) Borda Count based on accuracy and Q-value of Q-statics for the selection by accuracy and diversity measure.

In a previous study [15], the seven DL architectures including VGG16, VGG19, ResNet 50, Inception V3, Inception ResNet V2, Xception, and MobileNet V2 were evaluated and compared using the BreakHis dataset. The main results have showed that the DL end-to-end architecture Xception achieved the highest accuracy using the 40X MF with a value of 90.91%, the DL end-to-end architecture VGG16 achieved 90.43% on 100X MF, and the DL end-to-end architecture MobileNet V2 performed best on 200X and 400X with accuracy values of 90.90% and 89.91%, respectively. Furthermore, it is noticeable that the DL model ResNet 50 underperformed compared to the other architectures regardless the MF used and therefore it was considered as a baseline.

Table 3 The ranking of the seven DL models to design the DEHtE based on the selection by accuracy strategy

Rank	40X	100X	200X	400X
1	XCEP	VGG16	MOBN	MOBN
2	MOBN	MOBN	XCEP	INRES
3	INRES	XCEP	VGG16	VGG16
4	VGG16	INV3	VGG19	XCEP
5	INV3	INRES	INRES	INV3
6	VGG19	VGG19	INV3	VGG19
7	RES50	RES50	RES50	RES50

Table 4 Results in term of accuracy of the DEHtE designed using the selection by accuracy over the BreakHis dataset

DEHtE	40X	100X	200X	400X
E2HVA	90.9%	90.4%	90.9%	89.9%
E3HVA	93.2%	92.5%	92.8%	91.3%
E4HVA	93.3%	92.5%	92.7%	91.4%
E5HVA	93.4%	91.8%	93.0%	91.3%
E6HVA	93.5%	92.8%	93.0%	91.5%
E7HVA	93.0%	92.1%	93.1%	91.4%
E2WVA	92.0%	93.0%	92.3%	90.9%
E3WVA	93.2%	92.7%	92.9%	91.5%
E4WVA	93.5%	93.4%	93.0%	91.8%
E5WVA	93.7%	92.9%	93.2%	91.4%
E6WVA	93.8%	93.0%	93.2%	91.6%
E7WVA	93.8%	93.1%	93.3%	91.8%

The best accuracy values for each MF are in bold

a. Selectin by accuracy

To assess and compare the performances of DEHtE constructed based on the selection by accuracy strategy and combined using two voting techniques (majority and weighted voting), the seven DL architectures based on accuracy were ranked and then combined using majority and weighted voting following the ranking presented in Table 3. As results we obtained DEHtE of Top 2, 3, 4, 5, 6 and 7 DL models as base learners.

Table 4 indicates the accuracy values of the DEHtE selected by accuracy using majority and weighted voting methods over the BreakHis dataset four MFs. It is revealed that:

- For 40X MF, the foremost accuracy value obtained using majority voting was 93.5% reached by ensemble of size six, and the best accuracy obtained using weighted voting was 93.8% reached by ensembles of size six and seven. Moreover, ensemble of size two shows the worst accuracy value in both voting methods: majority and weighted voting with values: 90.9% and 92%, respectively.
- For 100X MF, the best accuracy value obtained using majority voting was 92.8% reached by ensemble of size six, and the best accuracy value obtained using weighted voting was 93.4% reached by ensemble of size four. Moreover, the worst accuracy value obtained using majority voting was 90.4% reached by ensemble of size two, and the

worst accuracy value obtained using weighted voting was 92.5% reached by ensemble of size three.

- For 200X MF, ensemble of size seven shows the best accuracy value in both voting methods: majority and weighted voting with values: 93.1% and 93.3%, respectively. Moreover, ensemble of size two shows the worst accuracy value in both voting methods: majority and weighted voting with values: 90.9% and 92.3%, respectively.
- For 400X MF, the best accuracy value obtained using majority voting was 91.5% reached by ensemble of size six, and the best accuracy obtained using weighted voting was 91.8% reached by ensembles of size four and seven. Moreover, ensemble of size two shows the worst accuracy value in both voting methods: majority and weighted voting with values: 89.9% and 90.9%, respectively.

In order to determine whether the DEHtE outperform their singles, we clustered the constructed ensembles following the selection by accuracy strategy and using two voting methods (majority and weighted voting) with their seven DL models used as base learners. Figure 3 illustrates the outcomes of the SK statistical test. It is observed that:

- For 40X MF, 6 SK clusters were identified. The first cluster includes 10 ensembles out of 12 (all ensembles except E2WVA which belongs to the second cluster, and E2HVA which belongs to the third cluster with the DL model XCEP). The fourth cluster contains VGG16, INRES, MOBN and INV3. Moreover, the fifth and the last clusters contain VGG19 and ResNe50, respectively.
- For 100X MF, 6 SK clusters were identified. The first cluster involves 7 ensembles out of 12 (E7WVA, E6WVA, E5WVA, E4WVA, E3WVA, E2WVA, E6HVA). The second cluster contains 4 ensembles (E7HVA, E5HVA, E4HVA, and E3HVA). E2HVA belongs to the third cluster with VGG16 and MOBN, the fourth cluster contains XCEP and INV3. Moreover, INRES and VGG19 belong to the fifth cluster, and the last cluster contains RES50.

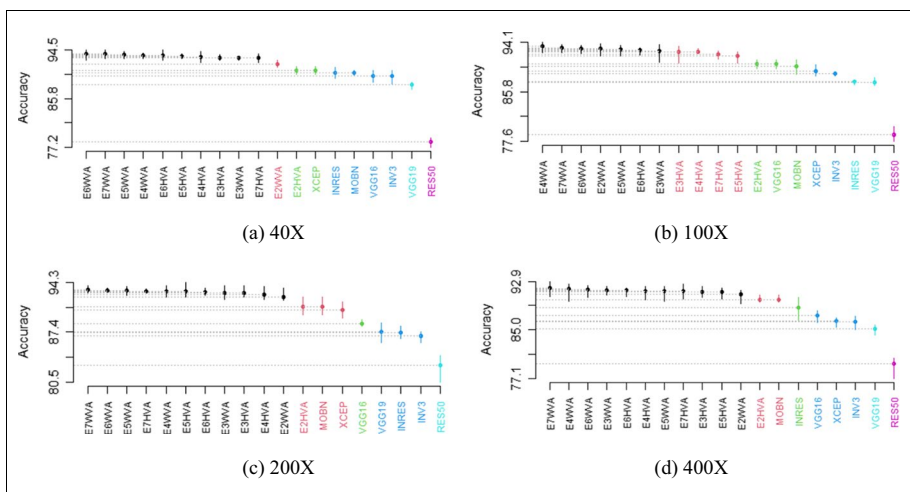


Fig. 3 SK test results of the DEHtE and their base learner designed using the selection by accuracy strategy over the BreakHis dataset

- For 200X MF, 5 SK clusters were obtained. The best SK cluster groups 11 ensembles out of 12 (all ensembles except E2HVA which was shown in the second cluster with MOBN and XCEP). The third cluster contains VGG16. The fourth cluster contains VGG19, INRES and INV3. Lately, RES50 was presented in the last cluster.
- For 400X MF, 6 SK clusters were obtained. The best SK cluster comprises 11 ensembles out of 12 (all ensembles except E2HVA which was shown in the second cluster with MOBN). The third cluster contains INRES. The fourth cluster contains VGG16, XCEP and INV3. The fifth and the last clusters contain VGG19 and ResNe50, respectively.

Table 5 shows the Borda Count ranking of the DEHtE selected by accuracy presented in the best SK test cluster over the BreakHis dataset. It is observed that:

- E7WVA was ranked the 1st on 40X, 200X and 400X MF, and 2nd on 100X MF.
- E6WVA was ranked 2nd on 40X and 200X, and 3rd on 100X and 400X.
- E5WVA was the 3rd on 40X and 200X, 5th on 100X, and 6th on 400X.
- E6HVA was ranked 4th on 40X and 200X MF, 5th on 400X, and 6th on 100X MF.
- E4WVA was ranked the 1st on 100X, 2nd on 400X, 4th on 40X, and 5th on 200X.
- E4HVA was 5th on 40X, 7th on 400X and 8th on 200X MF.
- E5HVA was the 5th on 200X, 6th on 40X, and 10th on 400X.
- E7HVA was ranked the 4th on 200X, 7th on 40X, and 8th on 400X MF.
- E3WVA was the 4th on 400X, 6th on 200X, 7th on 40X and 100X.
- E3HVA was ranked 7th on 200X, 9th on 400X, and 8th on 40X.
- E2WVA was ranked 4th on 100X, 9th on 200X, and 11th on 400X.

To sum up the obtained results, the SK test showed that:

- For 40X, 10 out of 12 ensembles gave significant results compared to their base learners (all ensembles except E2HVA and E2WVA).
- For 100X, 7 out of 12 ensembles gave significant results compared to their base learners (E7WVA, E6WVA, E5WVA, E6HVA, E4WVA, E3WVA, E2WVA).

Table 5 Borda count ranking of the DEHtE designed using the selection by accuracy strategy over the BreakHis dataset

DEHtE	40X	100X	200X	400X
E7WVA	1	2	1	1
E6WVA	2	3	2	3
E5WVA	3	5	3	6
E6HVA	4	6	4	5
E4WVA	4	1	5	2
E4HVA	5	-	8	7
E5HVA	6	-	5	10
E7HVA	7	-	4	8
E3WVA	7	7	6	4
E3HVA	8	-	7	9
E2WVA	-	4	9	11

Note that the first ranked proposed ensemble are in bold

- For 200X and 400X MFs, all ensembles except E2HVA gave significant results compared to singles.

The analysis above proves that the designed DEHtE significantly outperformed their base learners since they almost always belong to the best or second-best SK cluster. As results it is recommended to use the proposed approach to ameliorate the performances. Furthermore, the Borda Count method showed that **E7WVA** outperformed all other ensembles over the BreakHis dataset, since it was ranked top one on the three dataset MF (40X, 200X, and 400X) and second for the 100X MF.

b. Selection by diversity

To assess and compare the performances of DEHtE constructed based on the selection by diversity strategy and combined using two voting techniques (majority and weighted voting), we first calculate Q-statistic diversity of all combinations of sets of size two and select those that give the best Q-value. Once the best combination given the best Q-value is selected, we calculate Q-statistic of the previous set with the remaining DL technique (one by one) and select those that give best Q-value. The same process is repeated until reaching ensemble of size seven. The results of the ranked DL models using the Q-statistics diversity measure is represented in Table 6.

Table 7 shows the accuracy values of DEHtE selected by diversity using majority and weighted voting combination methods over the BreakHis dataset. It is noticed that:

- For 40X MF, ensemble of size seven achieved the best accuracy value using the majority and weighted voting methods with a value of 93.00% and 93.80%, respectively. Moreover, ensemble of size two achieved the worst accuracy value with both voting methods with a value of 90.9% and 91.4% respectively.
- For 100X MF, the best accuracy value obtained using majority voting was 92.7% reached by ensemble of size seven, and the best accuracy value obtained using weighted voting was 93.5% reached by ensemble of size six. Moreover, ensemble of size two achieved the worst accuracy value in both voting methods: majority and weighted voting with values: 90.1% and 90.3%, respectively.
- For 200X MF, the best accuracy value obtained using majority voting was 93.1% reached by ensemble of size five and seven, and the best accuracy obtained using weighted voting was 93.3% reached by ensemble of size seven. Moreover, ensemble of size two had the worst accuracy value in both voting methods: majority and weighted voting with values: 90.5% and 91.3%, respectively.
- For 400X MF, the best accuracy value obtained using majority voting was 91.4% reached by ensemble of size seven, and the best accuracy value obtained using weighted voting was 91.9% reached by ensembles of size five. Moreover, ensemble of size two had the worst accuracy value in both voting methods: majority and weighted voting with values: 86.4% and 86.8%, respectively.

Figure 4 illustrates the results of SK statistical test of ensembles designed using the selection by diversity strategy using majority and weighted voting methods over the BreakHis dataset. It is noticed that:

Table 6 Ranking results of the DL models used as base learners with the Q-statistic diversity measure over the BreakHis dataset

	Ranking of the DL models used as base learners	Mean Q-value diversity measure
40X	XCEP, RES50	0.557
	INV3, XCEP, RES50	0.621
	VGG16, INV3, XCEP, RES50	0.668
	MOBN, VGG16, INV3, XCEP, RES50	0.704
	VGG19, MOBN, VGG16, INV3, XCEP, RES50	0.731
	INRES, VGG19, MOBN, VGG16, INV3, XCEP, RES50	0.753
100X	MOBN, RES50	0.589
	INV3, MOBN, RES50	0.682
	XCP, INV3, MOBN, RES50	0.721
	VGG19, XCP, INV3, MOBN, RES50	0.727
	VGG19, VGG19, XCP, INV3, MOBN, RES50	0.746
	INRES, VGG19, VGG19, XCP, INV3, MOBN, RES50	0.761
200X	XCEP, RES50	0.562
	VGG19, XCEP, RES50	0.642
	MOBN, VGG19, XCEP, RES50	0.677
	VGG16, MOBN, VGG19, XCEP, RES50	0.693
	INV3, VGG16, MOBN, VGG19, XCEP, RES50	0.717
	INRES, INV3, VGG16, MOBN, VGG19, XCEP, RES50	0.737
400X	INV3, RES50	0.496
	VGG19, INV3, RES50	0.556
	INRES, VGG19, INV3, RES50	0.623
	XCP, INRES, VGG19, INV3, RES50	0.659
	VGG16, XCP, INRES, VGG19, INV3, RES50	0.681
	MOBN, VGG16, XCP, INRES, VGG19, INV3, RES50	0.707

Table 7 Results of the DEHtE in term of accuracy designed using the selected by diversity strategy over the BreakHis dataset

Ensembles	40X	100X	200X	400X
E2HVD	90.9%	90.1%	90.5%	86.4%
E3HVD	91.0%	90.7%	91.2%	88.9%
E4HVD	92.5%	91.4%	92.5%	89.9%
E5HVD	92.7%	91.8%	93.1%	91.1%
E6HVD	92.7%	92.7%	92.8%	90.9%
E7HVD	93.0%	92.1%	93.1%	91.4%
E2WVD	91.4%	90.3%	91.3%	86.8%
E3WVD	92.1%	91.9%	92.6%	89.2%
E4WVD	93.5%	91.8%	93.0%	90.9%
E5WVD	93.1%	93.3%	93.2%	91.9%
E6WVD	93.4%	93.5%	93.2%	91.4%
E7WVD	93.8%	93.1%	93.3%	91.8%

Note that the best accuracy values for each MF are in bold

- For 40X MF, we obtained 5 SK clusters. The first cluster contain 9 ensembles (all ensembles except E2WVD, E2HVD and E3HVD which were presented in the second cluster with XCEP). The third cluster contains INRES, MOBN, VGG16 and INV3. The fifth and the last clusters contain VGG19 and ResNe50, respectively.
- For 100X MF, we obtained 6 SK clusters. The first cluster contain 5 ensembles (E5WVD, E6WVD, E7WVD, E6HVD, and E7HVD). The second cluster contains 5 ensembles (E3WVD, E4WVD, E3HVD, E4HVD, and E5HVD). However, E2HVD was presented with VGG16 and MOBN in the third cluster, and the fourth cluster contains XCEP and INV3. Moreover, INRES and VGG19 were presented in the fifth cluster, and the last cluster contains RES50.
- For 200X MF, we obtained 5 SK clusters. The best SK cluster contain 9 ensembles (all ensembles except E2WVD, E2HVD, and E3HVD which were presented in the second cluster with MOBN and XCEP). The third cluster we observed VGG16, and the fourth cluster contains VGG19, INRES and INV3. The last cluster contains RES50.
- For 400X MF, we obtained 6 SK clusters. The first SK cluster contain 7 ensembles (E4WVD, E5WVD, E6WVD, E7WVD, E5HVD, E6HVD, and E7HVD). The second cluster contains E4HVD and MOBN. The third cluster contains E3WVD, E3HVD, and INRES. E2WVD and E2HVD were presented in the fourth cluster with VGG16, XCEP and INV3. The fifth and the last clusters contain VGG19 and ResNe50, respectively.

Table 8 shows the Borda Count ranking of the DEHtE designed using the selection by diversity strategy presented in the best SK test cluster over the four MF values. It is observed that:

- E7WVD was ranked the 1st on 40X and 200X, 2nd on 400X MF, and 3rd on 100X.
- E4WVD was ranked 2nd on 40X, 5th on 200X, and 6th on 400X.
- E6WVD was the 1st on 100X MF and the 3rd on 40X, 200X and 400X MF.

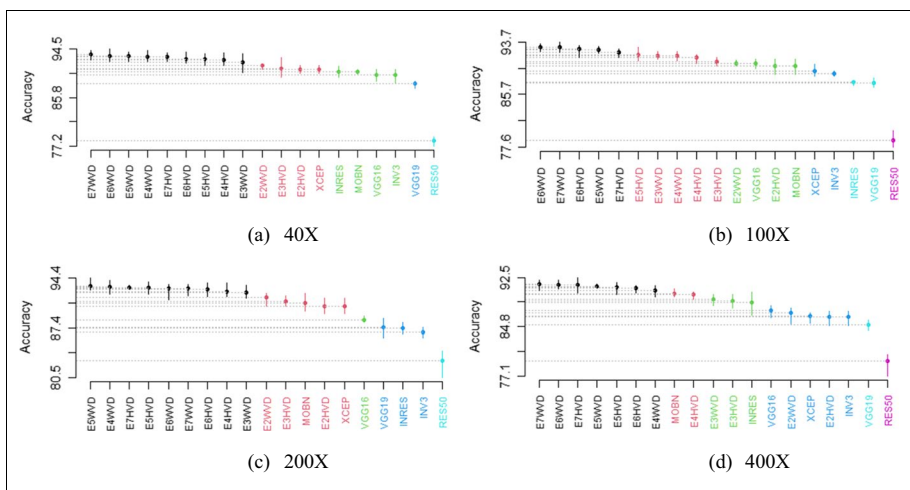


Fig. 4 SK test results of the DEHtE designed using the selection by diversity strategy over the BreakHis dataset

Table 8 Borda count ranking of the DEHtE designed using the selection by diversity over the BreakHis dataset

DEHtE	40X	100X	200X	400X
E7WVD	1	3	1	2
E4WVD	2	-	5	6
E6WVD	3	1	3	3
E5WVD	4	2	2	1
E7HVD	5	5	4	4
E5HVD	6	-	4	5
E6HVD	7	4	6	6
E4HVD	8	-	8	-
E3WVD	9	-	7	-

Note that the first ranked ensemble is in bold

- E5WVD was ranked the 1st on 400X MF, 2nd on 100X and 200X MF, and the 4th on 40X.
- E7HVA was ranked the 4th on 200X and 400X, and 5th on 40X and 100X MF.
- E5HVA was 6th on 40X, 4th on 200X, and 5th on 400X MF.
- E6HVA was the 4th on 100X, 6th on 200X and 400X, and 7th on 40X.
- E4HVA was ranked the 8th on 100X and 200X.
- E3WVA was the 7th on 100X, and the 9th on 100X.

To summarize the presented results, the SK test showed that:

- For 40X and 200X, 9 out of 12 ensembles gave significant results compared to their base learners (all ensembles except E2WVD, E2HVD and E3HVD).
- For 100X, 5 out 12 ensembles gave significant results compared to their base learners (E7WVD, E6WVD, E5WVD, E7HVD, and E6HVD).
- For 400X, 7 out of 12 ensembles gave significant results compared to their base learners (E7WVD, E6WVD, E5WVD, E4WVD, E7HVD, E6HVD and E5HVD).

Furthermore, the Borda Count ranking method showed that **E7WVD** outperformed all other DEHtE over the BreakHis dataset, since it was ranked top one on two MFs (40X, and 100X), second on 400X MF and third on 100X MF.

c. Selection by accuracy and diversity

To assess and compare the performances of DEHtE constructed based on the selection by accuracy and diversity strategy and combined using two voting rules (majority and weighted voting), we rank the DL models based on accuracy and pairwise Q-value calculated using Q-statics diversity measure. The results of the ranked DL models using accuracy and Q-statistics are represented in Table 9.

Table 10 shows the accuracy values of the DEHtE designed using the selection by accuracy and diversity strategy using majority and weighted voting methods. It is observed that:

- For 40X MF, the top accuracy value reached using majority voting was 93.2% reached by ensemble of size six, and the best accuracy obtained using weighted voting was

Table 9 Borda count ranking of the DEHtE designed using the selection by accuracy and diversity strategy over the BreakHis dataset

	Rank	DL models used as base learners
40X	1	XCP
	2	MOBN
	3	INV3
	3	VGG16
	5	RES50
	6	INRES
	7	VGG19
100X	1	MOBN
	2	VGG16
	3	INV3
	3	XCP
	5	RES50
	6	VGG19
	6	INRES
200X	1	XCP
	2	MOBN
	3	VGG19
	4	VGG16
	5	RES50
	6	INV3
	6	INRES
400X	1	INV3
	1	INRES
	3	MOBN
	5	RES50
	5	VGG19
	5	XCP
	5	VGG16

Table 10 Accuracy values of the DEHtE designed using the selection by accuracy and diversity over the BreakHis dataset

DEHtE	40X	100X	200X	400X
E2HVAD	90.9%	90.1%	90.5%	86.4%
E3HVAD	93.0%	92.2%	92.7%	90.7%
E4HVAD	93.0%	92.2%	92.8%	90.4%
E5HVAD	92.7%	91.8%	93.1%	91.0%
E6HVAD	93.2%	92.7%	92.8%	90.9%
E7HVAD	93.0%	92.1%	93.1%	91.4%
E2WVAD	92.0%	93.0%	92.3%	89.3%
E3WVAD	92.8%	92.5%	92.9%	90.9%
E4WVAD	93.2%	93.4%	93.0%	90.7%
E5WVAD	93.1%	93.3%	93.2%	91.1%
E6WVAD	93.6%	93.5%	93.2%	91.3%
E7WVAD	93.8%	93.1%	93.3%	91.8%

Note that the best accuracy values for each MF are in bold

- For 100X MF, we had 5 SK clusters. The first cluster contain 11 ensembles (all ensembles except the E2HVAD which was shown in the second cluster with XCEP and MOBN). Moreover, the third cluster contains VGG16. The fourth cluster contains VGG19, INRES and INV3. Lastly, RES50 was presented in the last cluster.
- For 200X MF, we had 6 SK clusters. The first SK cluster contains 5 ensembles (E4WVAD, E5WVAD, E6WVAD, E7WVAD, and E6HVAD). The second cluster contains 6 ensembles (E3HVAD, E4HVAD, E5HVAD, E7HVAD, E2WVAD, and E3WVAD). E2HVAD was presented in the third cluster with VGG16 and MOBN. The fourth cluster contains XCEP and INV3. The fifth cluster contains INRES and VGG19. Lastly, RES50 was presented in the last cluster.
- For 400X MF, we had 6 SK clusters. The first SK cluster contains MOBN and 10 ensembles (all ensembles except E2WVAD which was presented in the second cluster with INRES, and E2HVAD which was presented in the third cluster with VGG16, XCEP and INV3). The fourth and the last clusters contain VGG19 and ResNe50, respectively.

Table 11 shows the borda count ranking of ensembles selected by accuracy and diversity presented in the best SK test cluster over the four MF values. It is observed that:

- E7WVAD was ranked the 1st one on 40X, 200X and 400X MF, and 4th on 100X MF.
- E6WVAD was ranked the 1st on 100X, 2nd on 40X, and 3rd on 200X and 400X.
- E6HVAD was the 3rd on 40X, 5th on 200X and 400X, and 6th on 100X.
- E4WVAD was ranked 2nd on 100X, 4th on 40X and 200X MF, and 8th on 400X MF.
- E5WVAD was ranked the 3rd one on 100X, 4th on 400X, and 5th on 40X and 200X.
- E7HVAD was 2nd on 400X, 6th on 40X and 8th on 100X MF.
- E3HVAD was the 7th on 40X and 400X, and 10th on 100X MF.
- E4HVAD was ranked the 7th on 40X, and 9th on 100X and 400X MF.
- E3WVAD was the 6th on 400X, 7th on 100X, and 8th on 40X MF.
- E5HVAD was the 6th on 400X, 9th on 40X, and 11th on 100X MF.
- E2WVAD was the 5th on 100X, and 10th on 40X MF.
- MOBN was ranked the last on 400X MF.

Table 11 Borda count ranking of the DEHtE designed using the selection by accuracy and diversity strategy over the BreakHis dataset

DEHtE	40X	100X	200X	400X
E7WVAD	1	4	1	1
E6WVAD	2	1	3	3
E6HVAD	3	6	5	5
E4WVAD	4	2	4	8
E5WVAD	5	3	5	4
E7HVAD	6	8	-	2
E3HVAD	7	10	-	7
E4HVAD	7	9	-	9
E3WVAD	8	7	-	6
E5HVAD	9	11	-	6
E2WVAD	10	5	-	-
MOBN	-	-	-	10

Note that the first ranked ensemble is in bold

To summarize, the SK test showed that:

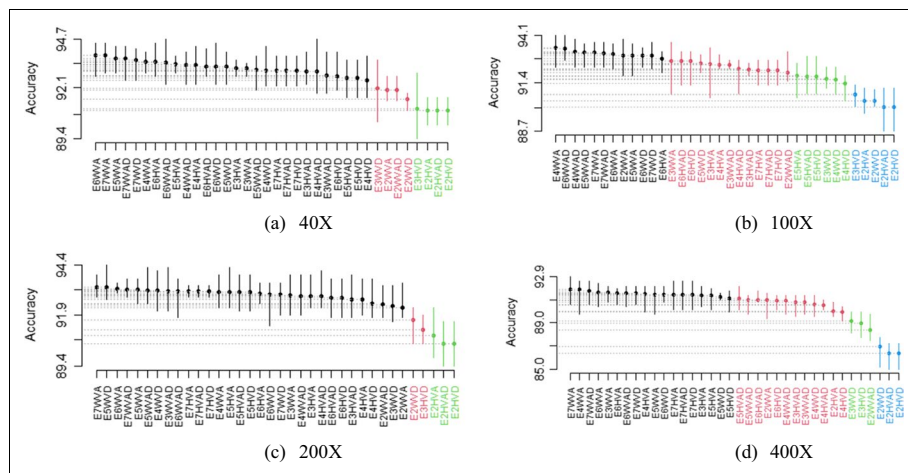
- For 40X and 100X, all ensembles except E2HVAD gave significant results compared to singles.
- For 200X, 5 out of 12 ensembles gave significant results compared to singles (E7WVAD, E6WVAD, E5WVAD, E4WVAD, and E6HVAD).
- For 400X, it is observed that the best cluster contain 10 ensembles and MOBN, thus ensembles in this MF are not significantly better than MOBN as single.

In addition to that, the Borda Count method showed that **E7WVAD** outperformed all other ensembles over the BreakHis dataset, since it was ranked top one on three dataset MF (40X, 200X, and 400X), and fourth on 100X.

6.2 Comparing the DEHtE regardless the selection strategy and voting method

This section compares the three selection strategies (by accuracy, by diversity and by both accuracy and diversity) used to construct the DEHtE ensembles and the two combination methods (majority and weighted voting methods) over the four MFs of the BreakHis dataset (40X, 100X, 200X and 400X). To compare between the three selection strategies and the two combination methods, we used (1) SK statistical test to cluster ensembles with no significant difference based on accuracy (Fig. 6), (2) the Borda Count voting method to rank ensembles obtained in the best SK test cluster based on the four metrics: accuracy, precision, recall and f1-score (Table 11) and (3) comparison of the best designed DEHtE with ensembles proposed in the literature.

Figure 6 illustrates the SK test of all ensembles constructed with the three strategies (SbA, SbD, and SbAD) and combined with the two voting methods (majority and weighted voting) over the BreakHis dataset. It is noticed that:



- For 40X MF, 3 SK clusters were obtained. The first cluster contains 28 out of 36 ensembles (all ensembles except E2HVA, E2HVD, E2HVAD, E2WVA, E2WVD, E2WVAD, E3HVD, and E3WVD). The second cluster contains four ensembles (E2WVA, E2WVD, E2WVAD, E3WVD). The last cluster shows 4 ensembles (E2HVA, E2HVD, E2HVAD, E3HVD).
- For 100X MF, 4 SK clusters were obtained. The first cluster contains 12 ensembles (E2WVA, E4WVA, E5WVA, E6WVA, E7WVA, E6HVA, E4WVAD, E5WVAD, E6WVAD, E7WVAD, E6WVD, E7WVD). The second cluster shows 13 ensembles (E3WVA, E3HVA, E4HVA, E7HVA, E3HVAD, E4HVAD, E6HVAD, E7HVAD, E2WVAD, E3WVAD, E6HVD, E7HVD, E5WVD). The third cluster shows 6 ensembles (E4HVD, E5HVD, E3WVD, E4WVD, E5HVA, E5HVAD). The last cluster contains 5 ensembles (E2HVA, E2HVD, E2WVD, E2HVAD, E3HVD).
- For 200X MF, we obtained 3 SK clusters. The best cluster contains 31 ensembles out of 36 (all ensembles except E2WVD, E3HVD, E2HVA, E2HVD, E2HVAD). The second cluster contains E2WVD, E3HVD, and the last cluster contains E2HVA, E2HVD, E2HVAD.
- For 400X MF, 4 SK clusters were obtained. The first cluster contains 18 ensembles (E3HVA, E3WVA, E4HVA, E4WVA, E5HVA, E5WVA, E6HVA, E6WVA, E7HVA, E7WVA, E7HVAD, E6WVAD, E7WVAD, E5HVD, E7HVD, E5WVD, E6WVD, E7WVD). The second cluster shows 12 ensembles (E2WVA, E2HVA, E3HVAD, E4HVAD, E5HVAD, E6HVAD, E3WVAD, E4WVAD, E5WVAD, E4HVD, E6HVD, E4WVD). For the third cluster, it contains 3 ensembles (E3HVD, E3WVD, E2WVAD). The last cluster shows 3 ensembles (E2HVD, E2WVD, E2HVAD).

As shown in the figure above, DEHtE designed using the SbA appeared more than the ones designed using other strategies in the best cluster for 100X and 400X MF (6 ensembles in 100X MF and 10 in 400X). As for the MFs 40X and 200X, DEHtE designed using the SbA and by the selection by the SbAD, appeared equally in the best cluster (10 ensembles in 40X MF, and 11 in 200X MF).

To select the best selection strategy and the best combination method to design the DEHtE, Table 12 shows the top five ranked ensembles of the best SK cluster over the four MF values of the BreakHis dataset using Borda Count voting method base on accuracy, precision, recall and F1-score.

It is observed in Table 11 that the DEHtE designed using majority voting don't appear on the top 5 ranks, regardless the selection strategy and MF value. Therefore, we can conclude that the weighted voting combination method provides better performing DEHtE compared to the majority voting method.

As for the DEHtE belonging to the top 5 ranks, we will focus the analysis on the ones ranked first regardless the MF. As result:

- E7WVD is ranked 1st on MF 40X and 200X, 2nd on 400X and 4th on 100X.
- E7WVA was ranked 1st on MF 40X and 200X, 2nd on 400X and 4th on 100X.
- E7WVAD was ranked 1st on MF 40X and 200X, 2nd on 400X and 4th on 100X.
- E6WVAD was the 1st on 100X MF, and 4th on 40X MF. It doesn't belong to the top 5 for the remaining MFs.
- E6WVD is ranked first for the MF 100X, and 5th for the MF 200X. It doesn't belong to the top 5-ranks for the remaining MFs.
- E4WVA is ranked first for the MF 400X and 2nd for the MF 100X. It doesn't belong to the remaining MFs.

Table 12 Top five ranked DEHtE belonging to the first SK cluster over the BreakHis dataset. Note that the bold entries represent the first ranked designed ensembles

40X		100X		200X		400X	
DEHtE	Rank	DEHtE	Rank	DEHtE	Rank	DEHtE	Rank
E7WVD	1	E6WVAD	1	E7WVD	1	E4WVA	1
E7WVA	1	E6WVD	1	E7WVAD	1	E7WVA	2
E7WVAD	1	E4WVA	2	E7WVA	1	E7WVAD	2
E6WVA	2	E4WVAD	2	E6WVA	2	E7WVD	2
E5WVA	3	E5WVAD	3	E5WVD	3	E5WVD	3
E6WVAD	4	E7WVA	4	E5WVAD	3	E6WVA	4
E4WVD	5	E7WVAD	4	E5WVA	4	E3WVA	5
-	-	E7WVD	4	E6WVD	5	-	-
-	-	E2WVA	5	-	-	-	-

To summarize the results obtained with SK test and Borda Count, and to analyze the impact of the selection strategy used to design the most outperforming DEHtE ensemble for the histological BC classification regardless the MF value, we count the number of occurrences of the DEHtE in the best SK cluster, and appearance in top-5 ranks given by Borda Count ranking.

As can be seen in Table 13, it is noticeable that the DEHtE designed using the selection by accuracy strategy outperform the others using the selection by diversity or by both accuracy and diversity strategies since it appeared 21 times in the best cluster when using weighted voting and 16 times when using majority voting. Plus, it was ranked 13 times in the top-5 using Borda Count ranking method and was ranked 3 times the 1st and 4 times the 2nd. The following strategy was selection by accuracy and diversity: it appeared 17 times in the best SK cluster when using weighted voting and 11 times when using majority voting, 9 times in top-5 Borda count ranking, and was ranked 3 times the 1st and 2 times the 2nd. Finally, selection by diversity underperform all other strategies: it appeared 14 times in the best SK cluster when using weighted voting and 10 times when using majority voting, 9 times in top-5 ranks, and was ranked 3 times 1st and 1 time 2nd.

In order to validate our proposed approach, Table 13 shows a comparative analysis of the best proposed DEHtE designed using the SbA strategy, the weighted voting combination method and seven DL models as base learners (E7WVA) with recent scientific publications proposing new ML and DL approaches tested using the BreakHis dataset and published between 2022 and 2023 in indexed journals. As results, the proposed DEHtE (E7WV) have the highest accuracy for the majority of MFs, with MF 40X outperforming the other used MFs and other state of the art suggested approaches (Table 14).

From the outcomes of this empirical study, the best combination method to design the DEHtE is the weighted voting, and the best selection strategy is the selection by accuracy, followed by the selection by both accuracy and diversity. Furthermore, it is recommended to use seven base learners since almost all the first ranked DEHtE are designed using seven DL models as base learners. In addition to the performances of the proposed DEHtE (E7WVA), the computational time required to design it, was not noticeably longer than the one required to train a single DL architecture using transfer learning for fine-tuning run on a TPU processing unit with 8 cores, 35 GB of RAM, and a Linux-based OS provided

Table 13 Appearance in the best SK cluster and Top-5 Borda count ranking

Voting Method	Selection Strategies	Appearance in the best SK cluster	Appearance in top 5 ranks	Borda count Ranking
weighted voting	Selection by Accuracy	21	13	3 times 1st 4 times 2nd 1-time 3 ^{thrd} 3 times 4th 2 times 5th
	Selection by Accuracy and diversity	17	9	3 times 1st 2 times 2nd 2 times 3 ^{thrd} 2 times 4th
	Selection by Diversity	14	9	3 times 1st 1 time 2nd 2 times 3 ^{thrd} 1 times 4th 1 times 5th
Majority voting	Selection by Accuracy	16	-	-
	Selection by Accuracy and diversity	11	-	-
	Selection by Diversity	10	-	-

by Google in Colab Notebook. This is due to the fact that when we train a single DL architecture, we must train the model for both feature extraction and classification and store the classification predictions, whereas the computational time for combining the predictions of the generated DL models using majority or weighted voting is small (30s) because these combination strategies are simple and do not require computational power.

7 Threats of validity

Internal Validity: In this study, we used a fivefold cross validation technique commonly used in ML and DL projects to assess a model's ability to predict new data points[25]. Another internal threat is using transfer learning for fine tuning, this involves freezing all convolutional base layers with ImageNet weights. Freezing or tuning some convolutional layers may affect the performance of classifiers.

External validity: The external threat's aim is to see if the study's findings are applicable to other contexts[46]. Since this study used the most widely used public histological dataset with different variants of the magnification factors [33], we cannot generalize the results to all datasets of the same image type. As a consequence, it is essential to test this study on public or private datasets in order to confirm or refute the study's findings.

Construct validity: The construct validity seeks to provide an answer to the measurement validity question [47], or, more precisely, the reliability of the measurements chosen to assess the performance of the proposed techniques. As a result, this study employs four performance measures (accuracy, precision, recall, and F1-score), the SK test to cluster statistically indifferent models and the Borda count voting technique, which considers the four-evaluation metrics, ensures that no performance metric is favored over another.

Table 14 Comparison of the proposed approach with related studies tested on the BreakHis Dataset

Authors and year of publication	BreakHis MFs	Accuracy (%)	Recall (%)	Precision (%)	F1-score (%)	Average Accuracy (%)
[28], 2023	40X	92.20	94.00	80.10	-	91.27
	100X	90.60	85.00	86.50	-	
	200X	92.60	94.10	81.40	-	
	400X	89.70	84.20	83.70	-	
[43], 2022	40X	87.90	91.60	90.20	91.60	87.92
	100X	89.10	94.20	92.30	92.50	
	200X	88.90	93.10	89.00	91.80	
	400X	85.80	92.30	87.50	89.80	
[44], 2022	40X	92.52	95.36	90.70	92.93	91.19
	100X	90.64	93.47	88.31	90.89	
	200X	92.01	94.94	89.75	92.26	
	400X	89.60	91.61	88.31	89.91	
[45], 2023	40X	93.32	-	-	-	92.14
	100X	93.67	-	-	-	
	200X	93.20	-	-	-	
	400X	88.40	-	-	-	
	100X					
	200X					
The proposed approach	40X	93.80	97.20	90.90	93.90	93.00%
	100X	93.10	96.90	90.10	93.40	
	200X	93.30	94.40	92.50	93.40	
	400X	91.80	95.70	89.10	92.20	

Additionally, this study did not take into consideration the execution time of the proposed ensembles when they were evaluated and compared.

8 Conclusion and future works

The binary classification of breast histopathological images using the BreakHis dataset is a problem that is addressed in this paper. It designed and proposed a deep end-to-end heterogenous ensemble learning approach based on seven DL models using fine-tuned VGG16, VGG19, RES50, INV3, INRESV2, XCEP, and MOBN. The proposed approach consists of using two voting methods (majority and weighted voting) and three selection strategies (accuracy, diversity, and both accuracy and diversity) to design deep end-to-end heterogenous ensembles of two up to seven DL. The evaluation techniques used to evaluate the proposed approach, were: four classification performance criteria (accuracy, precision, recall, and F1-score), SK statistical test, and Borda Count to assess and rank the proposed approach over the BreakHis dataset. The most important outcomes of this study are:

(RQ1): Does the DEHtE using voting combination methods outperform the DL models used as base learners?

The DEHtE developed using the two voting methods (majority and weighted voting) and the three selection strategies, showed a significant result compared to their base learners for all MFs, the accuracy value increased from 90.91% (the best accuracy value achieved on 40X by XCEP) to 93.8% achieved by E7WVA, E7WVAD and E7WVD. For 100X MF, the accuracy value increased by 3.07% (from 90.43% to 93.5% achieved by E6WAD and E6WVD). For 200X and, the accuracy value increased from 90.9% to 93.3% by the ensemble E7WVD, E7WVAD and E7WVA. For 400X MF, the accuracy increased from 89.91% to 91.8% by the ensemble E4WVD.

(RQ2): Which selection strategy in the design of the proposed approach (DEHtE) gives better results?

The results showed that DEHtE constructed using the selection by accuracy strategy outperform the other strategies, since they appeared 37 times regardless the voting method in the best SK test cluster, 13 times in top-5 Borda count ranking, and were ranked 3 times 1st and 4 times 2nd. The second ranked strategy is the selection by accuracy and diversity and finally the selection by diversity only.

(RQ3): Which voting combination method gives better results for the design of the DEHtE?

The outcomes of the analysis showed that no DEHtE designed using the majority voting combination rule appeared in the top 5-ranked ensembles using Borda Count in opposition with the weighted voting combination method. Thus, the weighted voting method is the best for all strategies and for all dataset MF.

Ongoing works intend to design ensembles using deep hybrid architectures as base learners in order to reduce the computation time. Moreover, we will investigate the explanation and interpretation of the different models used.

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Data availability The used dataset is the public histological dataset Breakhis dataset referenced by the reference [33] and cited in the threats of validity external validity line 2.

Code availability Not applicable.

Declarations

Conflict of interest All the authors declare that there is no conflict of interest regarding the publication of this paper.

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